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Editor-in-Chief

IEEE Transactions on Power Delivery

Dear Editor-in-Chief,

We submit for your consideration the original research article:

**“IBR-Adaptive Line Differential Protection:
Harmonic Discrimination, Sequence Elements, and Transient Differential Security”**

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Motivation and problem addressed. The progressive displacement of synchronous generators by inverter-based resources (IBRs) — solar PV, battery storage, and grid-forming wind converters — imposes a well-documented but unsolved problem on line differential (87L) protection. IBR fault-current limiters restrict the short-circuit contribution to 1.0–1.2 pu, producing a differential signal of 0.06–0.20 pu at the relay CT primary, which falls below the standard 87L pickup threshold. In our deterministic test matrix, the baseline relay fails on 9 of 21 scenarios (57% failure rate) at low IBR penetration. Three independent failure modes are responsible: magnitude suppression for three-phase faults, harmonic-induced confidence collapse, and negative-sequence suppression for asymmetric faults. No published 87L scheme addresses all three simultaneously, and no prior work has exploited the fault-inception transient as a primary detection feature for symmetric IBR faults.

Contributions. This paper proposes a five-element IBR-adaptive 87L protection chain that closes all three gaps:

1. **Physics-constrained adaptive threshold (C1):** The bias slope k_m^* is derived analytically from simultaneous CT-mismatch stability and high-impedance-fault sensitivity constraints as a closed-form function of the real-time IBR current ratio k_{ibr} . This eliminates the conservative fixed setting that previously caused either missed trips (low k_{ibr}) or false trips (high k_{ibr}).
2. **Frequency-adaptive harmonic pre-filter (C2):** A zero-crossing frequency tracker maintains the pi-section charging canceller orthogonal to 47–53 Hz; a QR-based harmonic regressor removes IBR-injected odd harmonics, restoring the relay confidence score from $c = 0.37$ to $c \geq 0.99$ under 10 % THD.
3. **Negative- and zero-sequence differential elements (C3):** An 87LN element provably achieves differential-to-bias ratio $\rho_2 = 2$ irrespective of k_{ibr} when the IBR suppresses its negative-sequence output — inverting the conventional assumption that IBR sequence suppression is always detrimental. An 87LN0 element extends coverage to solidly-earthed pure-IBR earth faults.
4. **Transient differential current signature element (C4):** TDCS-87L exploits the step-change in differential RMS at fault inception, which does not occur for external faults or load steps. An analytical bound proves detection for all faults with $k_{ibr} \geq 0.058$ pu. Monte Carlo Group C (5 000 external fault trials) confirms a 15.5 % security margin against Class 5P CT mismatch at the joint worst-case parameter corner.
5. **Solar PV two-parameter model and DDR selector (C5):** The zero-DC-offset property of GFL PV fault current is exploited to reduce the zone estimator from four parameters (condition number $\kappa_n \in [3, 36]$) to two ($\kappa_n = 1$ analytically). A DC Decay Ratio gate routes each relay cycle to the correct estimator, reducing the median PV-fault trip time from 63 ms to 42 ms.

Validation. The chain is validated deterministically on 30 scenarios (21 SG/IBR + 9 PV), with zero regression on the 10 synchronous-generator reference cases, and probabilistically on 20 000 Monte Carlo trials ($N = 5\,000$ per group, four groups). Groups A–C (three-phase internal, SLG internal, external faults) achieve TPR = 1.000 and FPR = 0.000 with 95 % Wilson CI lower bound ≥ 0.9992 . Group D (PV internal faults) achieves TPR = 0.9984 ($CI_{lo} \geq 0.9976$). All results exceed IEC 60255-181 performance targets.

Fit for *IEEE Transactions on Power Delivery*. The paper addresses a fundamental protection challenge that is directly relevant to the Transactions' readership: relay engineers and system planners dealing with the ongoing transition to high-IBR networks. The combination of closed-form analytical derivation, formal propositions with proofs, and large-scale Monte Carlo certification meets the technical depth expected of a Transactions paper. The work is original and has not been submitted elsewhere.

Suggested reviewers. Given the subject matter, we suggest the following independent experts (none have been involved in this work):

- Prof. Ali Hooshyar, University of Toronto (IBR protection, line differential relaying)
- Prof. Evangelos Farantatos, EPRI (IBR fault current characterisation, adaptive protection)
- Prof. Hossein Hooshyar, NTNU (IBR-impacted protection coordination)
- Dr. Bogdan Kasztenny, SEL (differential relay design and negative-sequence elements)

Authorship and conflicts. Both authors declare no conflicts of interest. The contributions statement is provided separately in the portal upload. All data and simulation parameters are disclosed in the paper body and referenced technical reports; simulation code is available upon request.

We look forward to the editorial review and are available to respond to any reviewer queries.

Yours sincerely,

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